<u>Effective mass of an electron</u>: when electron is accelerated inside the crystal; The mass of an electron inside the crystal appears in general, different from free electron mass and is usually referred as effective mass. m^*.
Since electrons in the crystal is refered as quantum particles. Therefore it will have the group velocity under the influence of electric field.

$$v_g = \frac{d\omega}{dk} \quad -(1)$$

The energy of the quantum particle is given by

$$E = \hbar\omega \quad -(2)$$

$$\Rightarrow \omega = \frac{E}{\hbar}$$

$$\Rightarrow \frac{d\omega}{dk} = \frac{1}{\hbar}\frac{dE}{dk}$$

Therefore equation (1) can be written as.

$$v_g = \frac{1}{\hbar}\left(\frac{dE}{dk}\right) \quad -(3)$$

~~Acceleration due to electric field~~
Acceleration can be written as

$$a = \frac{dv_g}{dt} = \frac{1}{\hbar}\frac{d}{dt}\left(\frac{dE}{dk}\right)$$

$$a = \frac{1}{\hbar}\frac{d^2E}{dk^2}\frac{dk}{dt} \quad -(4)$$

Momentum of the quantum particle like electron can be written as

$$p = \hbar k$$

$$\Rightarrow \frac{dp}{dt} = \hbar \frac{dk}{dt}$$

$$\Rightarrow \frac{dk}{dt} = +\frac{1}{\hbar} \frac{dp}{dt} \quad \text{— (5)}$$

From equations (4) & (5)

$$a = \frac{1}{\hbar^2}\left(\frac{d^2E}{dt^2}\right)\frac{dp}{dt}$$

$$\Rightarrow \frac{dp}{dt} = F_{ext} = \left(\frac{\hbar^2}{\frac{d^2E}{dk^2}}\right) a$$

$$\boxed{m^* = \frac{\hbar^2}{\frac{d^2E}{dk^2}}}$$

Fermi Energy : It is the energy difference between heighest and lowest ~~possible~~ occupied single particle states in a quantum system of noninteracting fermions (electrons) at absolute zero. It is represented by E_F.
$(T = 0°K)$

Fermi Function :→ This function gives the probability distribution ~~available~~ for the available electrons energy states will be occupied at a given Temperature T K.

$$f(E) = \frac{1}{\left(e^{\frac{E-E_F}{K_B T}} + 1\right)}$$

E_F = Fermi energy

K_B = Botzmann constant

T = Temperature in K.

Number of electrons per unit volume of the solid.

If n is the number of electrons per unit volume of the solid can be deduceced by

$$n = \int_0^{E_F} \{(\text{Density of states in the energy range } E \,\&\, (E+dE))\} \times (\text{Fermi distribution function})$$

$$n = \int_0^{E_F} g(E)\, dE \times f(E)$$

$$\boxed{n = \int_0^{E} \frac{8\pi\sqrt{2}\, m^{3/2}}{h^3} E^{1/2} \frac{dE}{\left(e^{\frac{E-E_F}{K_B T}}+1\right)}}$$

Number of electrons (n) at absolute zero ($T = 0\,°K$) & Fermi energy at $T = 0\,°K$

The number of electrons per unit volume can be written as.

$$n = \int_0^{E_F} g(E) f(E)\, dE \quad \text{—(1)}$$

③

$$\Rightarrow n = \frac{\pi 8\sqrt{2}\, m^{3/2}}{h^3} \int_0^{E_F} \frac{E^{1/2}\, dE}{\left(e^{\frac{E-E_F}{K_B T}} + 1\right)} \quad -②$$

Fermi function at $T = 0^\circ K$ & $E < E_F$

$$f(E) = \frac{1}{e^{\frac{(E - E_F)}{K_B T}} + 1}$$

$E - E_F = (-ive.)$ for $E < E_F$

$$\Rightarrow \frac{(E - E_F)}{K_B T} = \frac{(E - E_F)}{K_B \times 0} = \frac{-ive}{0} = -\infty$$

$$\Rightarrow f(E) = \frac{1}{e^{-\infty} + 1} = \frac{1}{\frac{1}{\infty} + 1} = 1$$

Now the equation ② can be written as.

$$n = \frac{\pi 8\sqrt{2}\, m^{3/2}}{h^3} \int_0^{E_F} E^{1/2}\, dE$$

$$= \frac{\pi 8\sqrt{2}\, m^{3/2}}{h^3} \frac{2}{3} E_F^{3/2}$$

$8 \times 2 \times \sqrt{2} = 8^{3/2}$

$$\boxed{n = \frac{\pi}{3} \left(\frac{8m}{h^2}\right)^{3/2} E_F^{3/2}} \text{ at } T = 0^\circ K$$

$$\boxed{E_F = \left(\frac{h^2}{8m}\right) \left(\frac{3}{\pi}\right)^{2/3} n^{2/3}} \text{ at } T = 0^\circ K$$

$E_F = 5.85 \times 10^{-38} \cdot n^{2/3}$ where $B = \frac{h^2}{8m}\left(\frac{3}{\pi}\right)^{2/3}$ Joules

$$\boxed{E_F = B n^{2/3}}$$

$$\cancel{B = \frac{h^2}{8m}\left(\frac{3}{\pi}\right)^{2/3}}$$

$$B = \frac{h^2}{8m}\left(\frac{3}{\pi}\right)^{2/3}$$

$$\boxed{B = 5.85 \times 10^{-38} \text{ Joules}}$$ for electron.